

3 (a) DNA and RNA are nucleic acids.

Figure 3.1 shows the five nitrogenous bases that are components of nucleic acids.

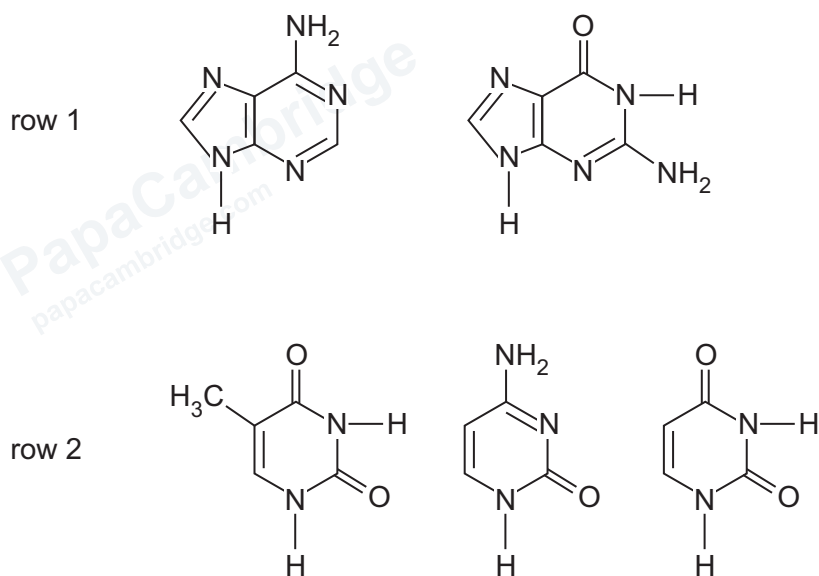


Figure 3.1

(i) A student made two correct statements about the nitrogenous bases shown in Figure 3.1:

- DNA and RNA have the nitrogenous bases in row 1
- DNA and RNA each have two of the nitrogenous bases in row 2, but **not** all three.

Explain how the structure of DNA and the structure of RNA support these two statements.

.....

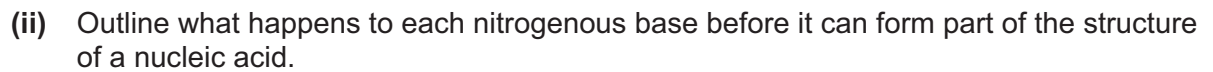
.....

.....

.....

.....

..... [3]



(iii) Describe the features of the genetic code that explain how a sequence of nitrogenous bases in messenger RNA (mRNA) can determine the sequence of amino acids in a polypeptide.

[4]

(b) Figure 3.2 shows the mitotic cell cycle of a mammalian stem cell.

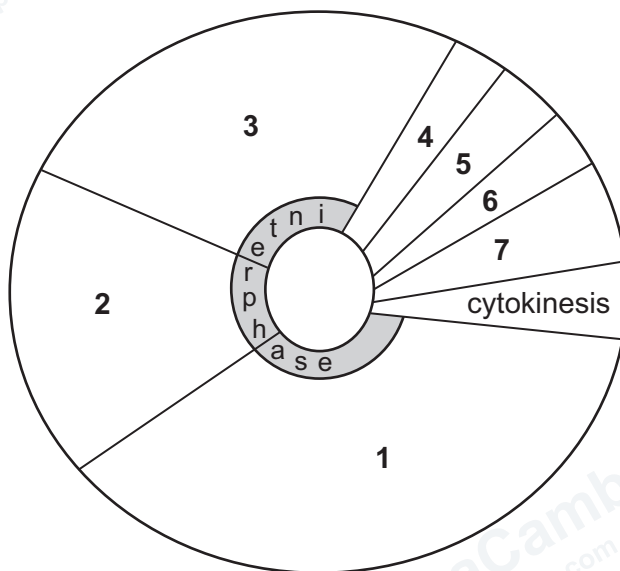


Figure 3.2

- (i) There are various processes involving DNA in the nucleus that occur during the cell cycle.

Complete Table 3.1 to identify the process involving DNA in the nucleus in each of the stages of the cell cycle, identified by the numbers on Figure 3.2. The process for stage 3 has been done for you.

Table 3.1

stage	process involving DNA in the nucleus
1	
2	
3	error checking
4	

[3]

- (ii) Describe how cytokinesis results in the formation of two separate cells.

.....

.....

.....

..... [2]



(iii) State the function of DNA polymerase **and** the function of DNA ligase in the cell cycle.

DNA polymerase

.....

.....

DNA ligase

.....

.....

[3]

[Total: 17]

PapaCambridge
papacambridge.com

Par
papa

PapaCambridge
papacambridge.com

PapaCambridge
papacambridge.com

PapaCambridge

PapaCambridge
bridge.com